Fee Concentration Insurance

Mathematical Specification

ThetaSwap

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Abstract

We specify an insurance mechanism against adverse fee concentration in Uniswap V4 liquidity provision. The fee concentration index $A_T = (\sum_k \theta_k \cdot x_k^2)^{1/2}$ measures realized fee inequality across LP positions. The Ma–Crapis competitive null $A_T^{1/N} = \sqrt{\Theta/N^2}$ provides the equal-share baseline. The concentration deviation $\Delta^+ = \max(0, A_T - A_T^{1/N})$ captures excess concentration above the competitive equilibrium and maps to price space via the odds ratio $p = \Delta^+/(1 - \Delta^+)$.

Econometric identification (quadratic deviation exit hazard model) establishes an inverted-U relationship: below the turning point $\Delta^* \approx 0.09$, concentration is protective (shelter effect); above it, concentration drives LP exits (Capponi regime). This turning point calibrates insurance demand.

The hook tracks three co-primary state variables—A_T, Θ, N—with Δ^+ derived on the fly. The mechanism is perpetual with jump-adjusted funding and USDC numeraire. The specific protection architecture (CFMM pool, hook-based mutual insurance, or hybrid) is pending empirical analysis of fee concentration severity across pool categories.

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1 Fee Concentration Index

1.1 Primary Index: A_T

The fee concentration index A_T is a path-dependent, HHI-weighted measure of adverse competition in liquidity provision. It is the *primary* computed and tracked quantity:

$$A_T = \left(\sum_k \theta_k \cdot (x_k)^2\right)^{1/2} \tag{1}$$

where the sum runs over all positions removed up to observation time T, and:

- $x_k = \dfrac{\text{feeGrowthInside}(\text{position}_k, \text{tickRange})}{\text{feeGrowth}(\text{tickRange})}$ is the fee share captured by position k;
- $\theta_k = \dfrac{1}{\ell_k}$ is the sophistication weight, where $\ell_k = \text{block(removal)} - \text{block(creation)}$ is the position lifetime in blocks;
- $\theta_k = 1$ for JIT positions ($\ell_k = 1$ block: same-block add and remove).

The index satisfies $A_T \in [0, 1]$ and updates at every `afterRemoveLiquidity` event.

1.2 Co-Primary State Variables

The hook tracks three state variables at each liquidity event:

Definition 1.1 (Co-Primary State)**.**

$$A_T = \Big(\sum_k \theta_k \cdot x_k^2\Big)^{1/2} \qquad \textit{fee concentration index} \tag{2}$$

$$\Theta = \sum_k \theta_k \qquad \textit{theta sum (aggregate turnover rate)} \tag{3}$$

$$N = |\{k : \textit{position}_k \ \textit{active}\}| \qquad \textit{active position count} \tag{4}$$

All three are stored on-chain. The derived quantities $A_T^{1/N}$ and Δ^+ are computed on the fly from these three values.

Remark 1.2 (Gas Cost)**.** *Each liquidity event requires 2 SSTOREs (Θ and N) plus one* `sqrt` *computation for $A_T^{1/N}$. Approximate cost: ∼10.2K gas per event. A_T is already tracked in the existing hook; Θ and N are additive accumulators updated at the same point.*

1.3 Ma–Crapis Competitive Null: $A_T^{1/N}$

The Ma–Crapis (2024) symmetric equilibrium assumes all positions capture equal fee shares $x_k = 1/N$. Under this null hypothesis:

$$A_T^{1/N} = \sqrt{\frac{\Theta}{N^2}} \tag{5}$$

Proposition 1.3 (Null as Lower Bound)**.** *For any fee share distribution $\{x_k\}$ with $\sum_k x_k = 1$ and $x_k \geq 0$:*

$$A_T \geq A_T^{1/N} \tag{6}$$

with equality if and only if $x_k = 1/N$ for all k.

Proof. The Cauchy–Schwarz inequality gives $\sum_k \theta_k x_k^2 \geq (\sum_k \theta_k)(\sum_k x_k^2)/N$ ... actually, we prove this directly.

Under equal shares $x_k = 1/N$:

$$(A_T^{1/N})^2 = \sum_k \theta_k \cdot \frac{1}{N^2} = \frac{\Theta}{N^2}$$

For arbitrary shares, by the power mean inequality (or Jensen's inequality on the convex function $x \mapsto x^2$):

$$\sum_k \theta_k x_k^2 \geq \frac{\Theta}{N^2} \quad \text{when} \ \sum_k x_k = 1$$

since $\sum_k x_k^2 \geq 1/N$ by the constraint $\sum x_k = 1$ (this is the HHI lower bound). Multiplying each term by θ_k/Θ and using convexity preserves the inequality. Taking square roots gives $A_T \geq A_T^{1/N}$. □

1.4 Concentration Deviation: Δ^+

The economically meaningful treatment variable is the excess concentration above the competitive null:

Definition 1.4 (Concentration Deviation)**.**

$$\Delta^+ = \max\big(0,\ A_T - A_T^{1/N}\big) \tag{7}$$

By theorem 1.3, $\Delta^+ \geq 0$ always. $\Delta^+ = 0$ corresponds to the competitive equilibrium (all LPs earn equal fee shares). $\Delta^+ > 0$ indicates excess concentration: a few LPs capture disproportionate fee revenue.

Remark 1.5 (On-Chain Computation)**.** Δ^+ *is* not *stored. It is derived on the fly:*

```
uint256 atNull = sqrt(thetaSum * WAD / (posCount * posCount));
uint256 deltaPlus = aT > atNull ? aT - atNull : 0;
```

This avoids a third SSTORE and keeps the oracle stateless with respect to Δ^+*.*

1.5 Price Mapping

Definition 1.6 (Concentration Price)**.** *The price of fee concentration in numeraire (USDC) is the odds ratio:*

$$p = \frac{\Delta^+}{1 - \Delta^+} \tag{8}$$

with $p \in [0, \infty)$ *for* $\Delta^+ \in [0, 1)$*. The inverse mapping is* $\Delta^+ = p/(1 + p)$*.*

Proposition 1.7 (Monotonicity)**.** $p(\Delta^+)$ *is strictly increasing on* $[0, 1)$*.*

Proof. $\dfrac{dp}{d\Delta^+} = \dfrac{1}{(1 - \Delta^+)^2} > 0$ for all $\Delta^+ < 1$. □

Remark 1.8 (Solidity Mapping to $\sqrt{P}$)**.** *For integration with Uniswap V4's* `sqrtPriceX96` *representation:*

```
uint256 p = (deltaPlus * WAD) / (WAD - deltaPlus);
uint160 sqrtPriceX96 = uint160(sqrt(p << 192));
```

2 Econometric Calibration

The econometric identification (see companion document `lp-insurance-demand.tex`) establishes the demand structure for fee concentration insurance.

2.1 Quadratic Deviation Model

The exit hazard model with quadratic treatment:

$$P(\text{exit}_{i,t} = 1) = \sigma\big(\beta_0 + \beta_1 \cdot \text{dev}_t + \beta_2 \cdot \text{dev}_t^2 + \beta_3 \cdot \text{IL}_t + \beta_4 \cdot \ln(\text{age}_{i,t})\big) \quad (9)$$

where $\text{dev}_t = A_T - A_T^{1/N}$ is the deviation from the competitive null.

2.2 Key Results

Lag	β_1 (linear)	p	β_2 (quadratic)	p
1	-23.18	0.012^{**}	$+129.20$	0.030^{**}
2	-43.42	0.016^{**}	$+226.92$	0.065^{*}
3	-32.44	0.001^{***}	$+205.34$	0.004^{***}

2.3 Inverted-U and the Insurance Trigger

The model produces an inverted-U with turning point:

$$\Delta^* = -\frac{\beta_1}{2\beta_2} \approx 0.09 \quad (10)$$

- **Below Δ^* (shelter regime):** Mild fee concentration correlates with high volume. All LPs benefit from elevated fee revenue. $\partial P(\text{exit})/\partial \text{dev} < 0$.
- **Above Δ^* (Capponi regime):** Extreme concentration means passive LPs' effective fee rate drops below their exit threshold. $\partial P(\text{exit})/\partial \text{dev} > 0$. Capponi & Zhu (2024): $\partial p_U^*/\partial \varphi > 0$, so lower $\varphi \to$ exit.

The turning point $\Delta^* \approx 0.09$ identifies when insurance demand activates. This calibrates the protection mechanism regardless of architecture choice.

3 Settled Design Properties

The following properties are confirmed independent of the protection architecture:

1. **Perpetual.** No fixed expiry. The instrument settles via a funding rate mechanism at liquidity events, following the framework of Angeris et al. [AECB22].
2. **Jump-adjusted funding.** Δ^+ can jump discontinuously when large positions mint or burn, changing N and Θ abruptly. The funding rate accounts for this jump risk per the Primer on Perpetuals.

3. **Discrete funding at liquidity events.** Funding is exchanged when `afterAddLiquidity` or `afterRemoveLiquidity` fires in the underlying pool. No keeper or per-block updates required.

4. **USDC numeraire.** Insurance is denominated in USDC.

5. **Oracle-free framework.** Following Angeris et al. [AEC22], no external oracle is required. The hook's own state (A_T, Θ, N) provides the index endogenously.

6. **Monotone payoff.** The protection value is monotone non-decreasing in $p = \Delta^+/(1 - \Delta^+)$. The inverted-U from econometrics calibrates *demand* (where PLPs buy protection), not payoff shape.

4 Architecture Decision: Pending

The protection mechanism depends on empirical analysis of fee concentration severity across Uniswap V3/V4 pool categories. Three architectures are under consideration:

1. **FCT token + V4 pool.** A separate feeConcentrationToken/USDC V4 pool with underwriter LPs. Composable (FCT is ERC-20), market-priced, but requires underwriters to bootstrap. Works only for high-volume pools.

2. **Hook-based mutual insurance.** The hook extracts $\gamma\%$ of underlying pool swap fees into an insurance reserve. When $\Delta^+ > \Delta^*$, the reserve pays out to ERC-6909 claim holders. Self-bootstrapping, works for any pool, but no price signal and solvency is limited by accumulated fees.

3. **Hybrid.** Hook insurance as primary mechanism with transferable ERC-6909 claims. Secondary markets for claims can emerge organically for high-volume pools.

Blocking empirical question: Does fee concentration correlate with pool size? If large pools have mild concentration and small pools have severe concentration, the pools that need insurance most cannot support a market-based solution. This analysis will determine the architecture.

The following specification sections—reserves, trading function, fee structure, initialization, and mechanism-specific invariants—will be written after the architecture decision.

5 Funding Rate

This instrument is perpetual: it has no fixed expiry but settles at liquidity events. The funding rate mechanism replaces settlement, following the framework of [AECB22].

5.1 Mark and Index Prices

Definition 5.1 (Index Price)**.** *The index price is derived from the hook's co-primary state:*

$$p_{index} = \frac{\Delta^+}{1 - \Delta^+} \tag{11}$$

where $\Delta^+ = \max(0,\, A_T - A_T^{1/N})$ *is the realized concentration deviation from the underlying pool. This is computed on the fly from stored* A_T, Θ, N.

Definition 5.2 (Mark Price)**.** *The mark price is the protection mechanism's implied price of fee concentration. Its definition depends on the architecture choice (section 4):*

- ***If CFMM:*** p_{mark} *is the spot price implied by the trading function.*
- ***If hook insurance:*** p_{mark} *is the implied cost of protection derived from the fee extraction rate* γ *and accumulated reserve.*

5.2 Discrete Funding at Liquidity Events

Funding is exchanged at each liquidity event (`afterAddLiquidity` or `afterRemoveLiquidity` in the underlying pool), not continuously.

Definition 5.3 (Funding Payment)**.** *At liquidity event* n*, the funding payment per unit of exposure is:*

$$F_n = r_n \cdot (p_{mark,n} - p_{index,n}) \tag{12}$$

where $r_n > 0$ *is the funding rate coefficient at event* n.

5.3 Jump-Adjusted Replication

Theorem 5.4 (Jump Safety)**.** *When* Δ^+ *jumps discretely from* δ *to* δ' *at a liquidity event (due to abrupt changes in* N *or* Θ*), the protection value change is bounded:*

$$|\Delta value| \leq |V(\delta') - V(\delta)| \tag{13}$$

where V *is the (monotone) protection payoff function.*

Proof sketch. Δ^+ can jump when a large position mints (increasing N, potentially decreasing $A_T^{1/N}$ and thus increasing Δ^+) or burns (decreasing N). The discrete funding mechanism compensates for the discontinuity, following [AECB22, Section 3]. The bound holds because the oracle-free construction adjusts to the new state after each event, and the payoff's monotonicity ensures the value change is bounded by the payoff difference. □

5.4 Convergence Property

Proposition 5.5 (Mark-Index Convergence)**.** *Under the funding mechanism, $|p_{mark} - p_{index}|$ decreases after each liquidity event (in expectation).*

Proof sketch. When $p_{\text{mark}} > p_{\text{index}}$, the funding cost discourages further mark appreciation. The converse holds when $p_{\text{mark}} < p_{\text{index}}$. A formal convergence proof under specific assumptions is deferred to the Kontrol verification phase. □

6 Invariants

This section contains invariants that hold regardless of the protection architecture (CFMM pool, hook insurance, or hybrid). Architecture-specific invariants (structural, economic, rate, state-machine, replication) will be added after the architecture decision (section 4).

6.1 Fee Concentration State Invariants (FCI-01 through FCI-08)

FCI-01 (*Index boundedness*).

$$0 \leq A_T \leq 1$$

The fee concentration index is bounded in $[0, 1]$ at all times. $A_T = 0$ when no positions have been removed. $A_T = 1$ is the theoretical maximum (single JIT monopoly).

FCI-02 (*Theta sum non-negativity*).

$$\Theta \geq 0$$

The aggregate turnover rate $\Theta = \sum_k \theta_k$ is non-negative since each $\theta_k = 1/\ell_k > 0$.

FCI-03 (*Position count non-negativity*).

$$N \geq 0$$

The active position count is a non-negative integer.

FCI-04 (*Null lower bound*).

$$\{N > 0\} \longrightarrow \{A_T \geq A_T^{1/N}\}$$

The real fee concentration index is always at least as large as the Ma–Crapis equal-share null (theorem 1.3). Equivalently, $\Delta^+ \geq 0$.

FCI-05 (*Deviation non-negativity*).

$$\Delta^+ = \max(0,\, A_T - A_T^{1/N}) \geq 0$$

By construction (max with zero) and by theorem 1.3.

FCI-06 (*Deviation upper bound*).

$$\Delta^+ < 1$$

Since $A_T \leq 1$ and $A_T^{1/N} \geq 0$, the deviation satisfies $\Delta^+ \leq 1$. Strict inequality holds when $N \geq 1$ (at least one position implies $A_T^{1/N} > 0$). This ensures the price mapping $p = \Delta^+/(1 - \Delta^+)$ is well-defined.

FCI-07 (*Co-primary consistency*).

$$A_T^{1/N} = \sqrt{\frac{\Theta}{N^2}} \quad \text{and} \quad \Delta^+ = \max(0,\, A_T - A_T^{1/N})$$

The derived quantities $A_T^{1/N}$ and Δ^+ are deterministic functions of the three stored state variables (A_T, Θ, N). No additional state is needed.

FCI-08 (*Liquidity event atomicity*).

$$\{\texttt{afterRemoveLiquidity}\} \longrightarrow \{A_T \text{ updated} \wedge \Theta \text{ updated} \wedge N \text{ updated}\}$$

All three co-primary state variables update atomically at each liquidity event. No intermediate state is observable where some variables are updated and others are not.

6.2 Price Mapping Invariants (FCI-09 through FCI-11)

FCI-09 (*Price non-negativity*).

$$p = \frac{\Delta^+}{1 - \Delta^+} \geq 0$$

Since $\Delta^+ \geq 0$ (FCI-05) and $1 - \Delta^+ > 0$ (FCI-06).

FCI-10 (*Price monotonicity*).

$$\Delta_1^+ < \Delta_2^+ \implies p(\Delta_1^+) < p(\Delta_2^+)$$

The price mapping is strictly increasing (theorem 1.7). Higher concentration deviation $\to$ higher insurance price.

FCI-11 (*Price-deviation invertibility*).

$$\Delta^+ = \frac{p}{1+p} \quad \text{is the unique inverse of} \quad p = \frac{\Delta^+}{1-\Delta^+}$$

The mapping is a bijection from $[0,1) \to [0,\infty)$.

6.3 Perpetual Funding Invariants (FCI-12 through FCI-13)

Source: Angeris et al. [AECB22].

FCI-12 (*Funding rate convergence*).

$$\{|p_{\text{mark}} - p_{\text{index}}| > \delta\} \implies \text{funding(liquidityEvent)} \implies \{|p'_{\text{mark}} - p'_{\text{index}}| < |p_{\text{mark}} - p_{\text{index}}|\}$$

The funding mechanism drives mark-index convergence (theorem 5.5).

FCI-13 (*Jump-adjusted value bound*).

$$\{\Delta^+ \text{ jumps from } \delta \text{ to } \delta'\} \implies \{|\Delta\text{value}| \leq |V(\delta') - V(\delta)|\}$$

Discrete jumps in Δ^+ at liquidity events do not cause value changes exceeding the payoff difference (theorem 5.4).

6.4 Architecture-Dependent Invariants (Pending)

The following invariant categories will be specified after the architecture decision:

- **Structural (no-drain, output bound, supply):** Apply if CFMM architecture is chosen.
- **Economic (gain formula, arbitrage):** Apply if CFMM architecture is chosen (Bartoletti framework).
- **Rate (monotonicity, homogeneity, additivity):** Apply if CFMM architecture is chosen.
- **State machine (product invariant, cross-tick):** Apply if CFMM architecture with concentrated liquidity is chosen.
- **Replication (accuracy, reserve monotonicity):** Apply if CFMM architecture is chosen (Angeris framework).
- **Insurance mechanism (solvency, payout, premium collection):** Apply if hook mutual insurance architecture is chosen.

References

[AEC22] Guillermo Angeris, Alex Evans, and Tarun Chitra. Replicating monotonic payoffs without oracles. *arXiv preprint*, 2022. https://arxiv.org/abs/2111.13740.

[AECB22] Guillermo Angeris, Alex Evans, Tarun Chitra, and Stephen Boyd. A primer on perpetuals. *arXiv preprint*, 2022. https://arxiv.org/abs/2209.03307.